

Sergi Sanchez Orvay

ROBOTICS RESEARCH INTERN AT BMW GROUP (ROBOTAC LAB) - MSC ROBOTICS STUDENT AT ETH ZÜRICH
Munich, Germany

☎ (+34) 610 47 84 53 | ✉ sergi.sanchez.orvay@gmail.com | 🏠 sergisanchezz.github.io | 💻 [sergisanchezz](https://sergisanchezz.com) | 🌐 [sergisanchezz](https://sergisanchezz.com)

Experience

Robotics Research Intern, [BMW Group \(RoboTac Lab\)](#) - Munich, Germany

SUPERVISOR: [PROF. DR. MOHSEN KABOLI](#)

Sept. 2025 - Present

Research at the intersection of embodied intelligence, cross-modal perception and robotic manipulation. Focusing on building robotic foundation models that unify tactile sensing, visual observations and action sequences into a shared representation using transformers and variational inference in PyTorch, enabling a pre-trained and generalizable model for diverse downstream tasks. Contributing in parallel to the [PHASTRAC](#) project (Horizon Europe), using Oscillatory Neural Networks to advance efficient edge intelligence for robotics.

Student Researcher, [ETH Zürich \(Vision for Robotics Lab\)](#) - Zürich, Switzerland

SUPERVISORS: [XINYI LI](#), [WILLIAM TALBOT](#), [DR. DAVID HUG](#), [DR. CORNELIUS VON EINEM](#), [PROF. DR. MARGARITA CHLI](#)

Feb. 2025 - July 2025

Advancing continuous-time SLAM through Gaussian Belief Propagation (GBP). Focused on analyzing and resolving the numerical instabilities of [Hyperion](#) (the first open-source continuous-time GBP solver) that prevented its deployability by identifying ill-conditioned landmark covariances and underconstrained spline tails as core failure modes. Designed and implemented improved regularization strategies, including diagonal and Levenberg-Marquardt regularization, message damping and spline tail-fixing, enabling stable on-line optimization. These improvements yielded millimeter-level trajectory accuracy, consistent convergence across synthetic and indoor sequences, and runtime reductions of more than 100x compared to [Ceres](#)-based Non-Linear-Least-Squares solvers, demonstrating the viability of GBP as an efficient continuous-time state estimation framework. Project report: [Gaussian Belief Propagation for Continuous-Time SLAM](#).

Robotics Research Intern, [Institut de Robòtica i Informàtica Industrial \(IRI-CSIC\)](#) - Barcelona, Spain

SUPERVISOR: [DR. JUAN ANDRADE-CETTO](#)

Sept. 2023 - July 2024

Research on lightweight, interpretable algorithms for motion and feature estimation using event cameras in an aerial robotic platform performing agile motion (accelerations of up to 40 rad/s² and 15 m/s²). Developed and implemented methods for event-based optical flow estimation using plane-fitting techniques on surfaces of active events. Implemented a state-of-the-art robust corner detection algorithm in Python. Designed a geometric model-based pipeline for egomotion estimation, combining normal flow, inverse depth optimization and RANSAC-based motion recovery using MATLAB and ROS2.

Education

MSc in Robotics, Systems and Control, [ETH Zürich](#) - Zürich, Switzerland

ACADEMIC SUPERVISOR: [PROF. DR. ROLAND SIEGWART](#)

Sept. 2024 - Present

Pursuing advanced training in robotics with a focus on embodied intelligence and robot learning & perception. Strong theoretical foundation in deep learning, robot control, probabilistic AI and computer vision, combined with extensive hands-on project experience. This includes work on [Gaussian Belief Propagation for Continuous-Time SLAM](#) with the [Vision for Robotics Lab](#), [Multi-Modal World Models for Autonomous Driving](#) with the [Computer Vision and Geometry Group](#) and ongoing research in cross-modal perception and tactile intelligence at the [BMW Group \(RoboTac Lab\)](#).

BSc in Electronic Eng. & Telecommunications, [Universitat Politècnica de Catalunya](#) - Barcelona, Spain

BACHELOR'S THESIS SUPERVISORS: [DR. JUAN ANDRADE-CETTO](#), [PROF. DR. MONTSE NÁJAR](#)

Sept. 2020 - July 2024

Ranked 1st of the class (GPA: 9/10) with honors in +30% of the courses and receiving the Best Academic Transcript Award. Conducted a research internship at [IRI-CSIC](#) working on vision algorithms for event-cameras that lead to my bachelor's thesis, [Event-based egomotion estimation](#) (Grade: 9.8/10).

Awards & Scholarships

Nova Talent Member (professional network connecting and mentoring top talent), [Nova](#) - Nov. 2024, Madrid, Spain

JAE Intro ICU - Robotics Research Scholarship, [Consejo Superior de Investigaciones Científicas \(CSIC\)](#) - 2023, Barcelona, Spain

Finalist in the Rohde & Schwarz International Engineering Competition, [Rohde & Schwarz](#) - 2023, Munich, Germany

Best student award in the Initial Phase of Electronic Eng. & Telecommunications degree, [UPC](#) - 2021, Barcelona Spain

Academic Excellence Award - Full financial support for the first university year, [Govern de les Illes Balears](#) - 2020, Ibiza, Spain